Practical - 14

Practical Title: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition: Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role. Record important admin actions in audit_logs.

A. Theory & Analysis

1. Are dashboard values loaded dynamically from the database?

Yes. Key statistics are calculated dynamically using SQL aggregation queries: SELECT COUNT(*) FROM students, SELECT COUNT(*) FROM events, SELECT COUNT(*) FROM registrations, and SELECT COUNT(*) FROM users. The numbers update automatically as records change.

2. Is access restricted to admin users?

Yes. Server-side middleware check_auth('admin') enforces role checks before rendering the page. Unauthenticated users are redirected to login.html, and authenticated students attempting to access admin files are redirected with an access denied alert.

3. Are role-based menus working?

Yes. The navigation bar dynamically adapts to the user's role: administrators receive administrative links (Overview, Manage Students, Manage Events, Events API CRUD), while students receive academic links.

4. Are important admin actions recorded in audit_logs?

Yes. The audit_logs table records security and management events (USER_LOGIN, CREATE_STUDENT, UPDATE_STUDENT, DELETE_STUDENT, CREATE_EVENT, UPDATE_EVENT, DELETE_EVENT) with user ID, action name, entity type, details, and timestamps, displayed in an activity log table on the admin dashboard.

B. Implementation Details

I built the admin dashboard in pages/admin_dashboard.php displaying live KPI metric cards, recent student enrollments, module quick links, and a System Audit Trail table reading from the audit_logs table. Access is restricted strictly to administrators using server-side session checks.

C. Evidence of Work

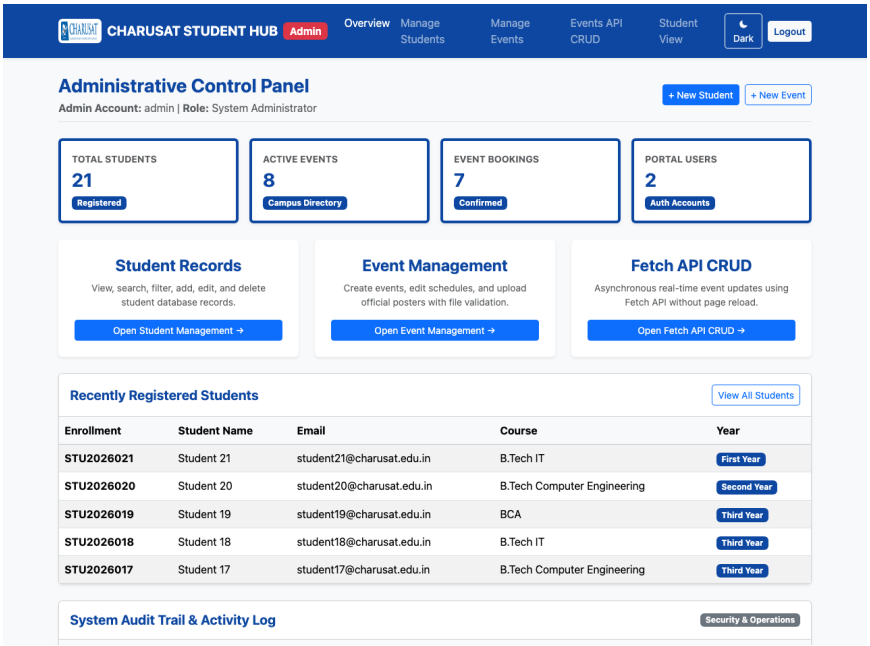


Figure: Admin Dashboard with Dynamic Metrics, Recent Enrollments, and System Audit Trail

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to develop a secure administrative dashboard featuring dynamic SQL KPI calculations, server-enforced role-based access control, contextual navigation menus, and comprehensive audit trail logging.